

Storage Management, LVM and Troubleshooting

Task 1 – Create and Mount an ext4 Partition

Question:

Use the Linux terminal to create a new partition on a virtual disk using the fdisk tool, format it as ext4, and mount it to /mnt/music.

Answer:

1. Identify the available disks using:

```
lsblk
```

2. Open the fdisk utility for the required virtual disk, for example:

```
sudo fdisk /dev/sdb
```

3. Create a new partition using the fdisk options for a new partition, then write the partition table.

4. Verify the new partition:

```
lsblk
```

5. Format the new partition as ext4:

```
sudo mkfs.ext4 /dev/sdb1
```

6. Create the mount directory:

```
sudo mkdir -p /mnt/music
```

7. Mount the partition:

```
sudo mount /dev/sdb1 /mnt/music
```

8. Verify the mount:

```
df -h
```

or

```
lsblk
```

Result:

The new virtual-disk partition is formatted with the ext4 filesystem and mounted successfully at /mnt/music.

Task 2 – Configure LVM

Question:

Set up an LVM group named `media_vg` with two physical volumes, create a logical volume `photos_lv` of 1GB, and mount it to `/mnt/photos`.

Answer:

1. Identify the two virtual disks or partitions using:

```
lsblk
```

2. Create physical volumes:

```
sudo pvcreate /dev/sdb1 /dev/sdc1
```

3. Create the volume group:

```
sudo vgcreate media_vg /dev/sdb1 /dev/sdc1
```

4. Create a 1GB logical volume:

```
sudo lvcreate -L 1G -n photos_lv media_vg
```

5. Format the logical volume as ext4:

```
sudo mkfs.ext4 /dev/media_vg/photos_lv
```

6. Create the mount directory:

```
sudo mkdir -p /mnt/photos
```

7. Mount the logical volume:

```
sudo mount /dev/media_vg/photos_lv /mnt/photos
```

8. Verify the configuration:

```
sudo pvs
```

```
sudo vgs
```

```
sudo lvs
```

```
df -h /mnt/photos
```

Result:

The two physical volumes are combined into the `media_vg` volume group, and the 1GB `photos_lv` logical volume is created, formatted, and mounted at `/mnt/photos`.

Task 3 – Simulate and Troubleshoot a Mount Failure

Question:

Simulate a hardware failure by unmounting a mounted partition and deleting its mount directory, then use Linux commands to diagnose and recover access to the data.

Answer:

1. Check the current mounts:

```
lsblk
```

```
mount
```

2. Unmount the partition:

```
sudo umount /mnt/music
```

3. Remove the mount directory to simulate the problem:

```
sudo rmdir /mnt/music
```

4. Check kernel/system messages for storage or filesystem information:

```
dmesg | tail -n 30
```

5. Recreate the mount directory:

```
sudo mkdir -p /mnt/music
```

6. Check the partition and filesystem:

```
lsblk
```

7. Mount the partition again:

```
sudo mount /dev/sdb1 /mnt/music
```

8. Verify that the data is accessible:

```
ls -la /mnt/music
```

Result:

The issue can be diagnosed by checking the block device with `lsblk`, mount status, and system messages with `dmesg`. Recreating the missing mount directory and mounting the correct partition restores access to the data.

Task 4 – Resize the Logical Volume

Question:

Resize the logical volume `photos_lv` to increase its size by 500MB, then verify the new size using the appropriate LVM and filesystem commands.

Answer:

1. Check the current logical-volume size:

```
sudo lvs
```

2. Increase the logical volume by 500MB:

```
sudo lvextend -L +500M /dev/media_vg/photos_lv
```

3. Resize the ext4 filesystem so it uses the additional space:

```
sudo resize2fs /dev/media_vg/photos_lv
```

4. Verify the logical-volume size:

```
sudo lvs
```

5. Verify the filesystem size:

```
df -h /mnt/photos
```

Result:

The `photos_lv` logical volume is increased by 500MB and the ext4 filesystem is expanded to use the newly allocated space.

Task 5 – Troubleshoot an Undetected USB Drive

Question:

A user reports that their USB drive is not detected when plugged in. List 3 Linux commands you would use to troubleshoot the issue and explain what each command checks.

Answer:

1. lsusb

This command lists USB devices detected by the system. It helps confirm whether the USB device is being recognized at the USB hardware level.

2. lsblk

This command lists block devices, including disks and partitions. It helps determine whether the USB drive has appeared as a storage device.

3. dmesg | tail -n 30

This command shows recent kernel messages. It can reveal USB connection errors, driver problems, filesystem issues, or other hardware-related messages.

Troubleshooting process:

First run lsusb immediately after connecting the drive. Then use lsblk to check whether a new storage device and partition have appeared. Finally, check dmesg for recent errors or warnings related to the USB device.

Result:

These three commands help determine whether the problem is related to USB detection, storage-device recognition, drivers, or another hardware/system issue.